

Final Exam Questions

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2. DAC: Ihzaz Abrar Sadik
3. Schmitt Trigger: Sazzadul Islam
4. Signal Generator: Shomen Kundu
5. NMOS+CMOS: Nur Mohammad Shaon

Please submit your questions by Sunday, 5 January 2025. Please make sure you make Set A and Set B of your question. Please provide a solution with the marks distribution to your question.

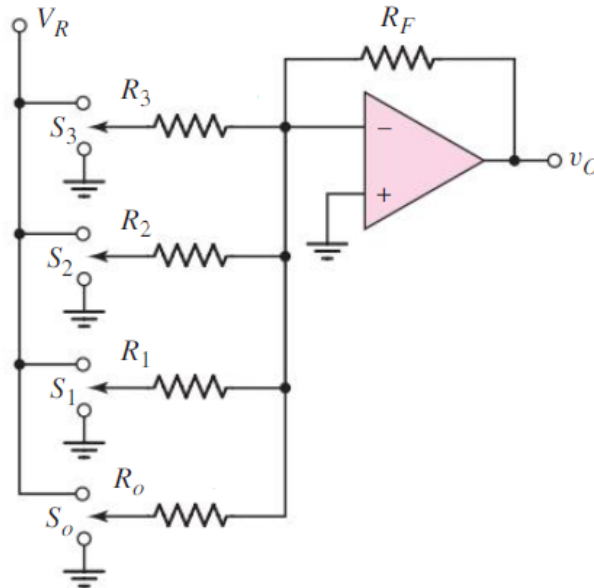
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- While posting your question here, please follow this format:

(a)	CO	Question	[Marks]
(b)	CO	Question	[Marks]
(c)	CO	Question	[Marks]

Question 2: DAC [ZAZ]

Set-A

For the binary weighted DAC shown below, answer the following questions:



Here, $V_R = -8V$, $R_3 = R_f$ & Input bit sequence: $S_3 S_2 S_1 S_0$

(a)	CO2	Show the output voltages for all possible input cases 'XXX0' in a table, where X may be 0 or 1.	[10]
(b)	CO2	What is the ratio of maximum output voltage & LSB(for input 0001) voltage?	[4]
(c)	CO2	What should be the ratio $\frac{R_3}{R_f}$, if we want the output voltage for input '1010' to be 5V?	[6]

Solution:

a)

Input	Output = $ V_R \times (B_3 \times 1 + B_2 \times 0.5 + B_1 \times 0.25)$
0000	0
0010	2
0100	4
0110	6
1000	8
1010	10
1100	12
1110	14

b) Maximum voltage = $|V_{ref}|(1 + 0.5 + 0.25 + 0.125) = 15V$

LSB voltage = $|V_{ref}|(0.125) = 1V$

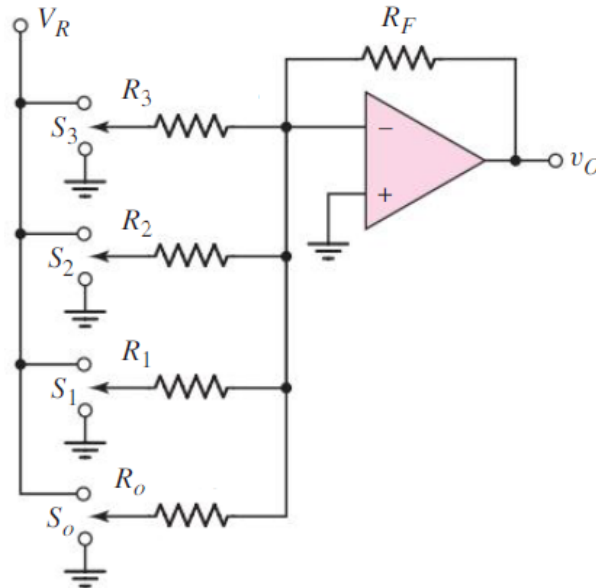
So, ratio = 15:1

c) Output voltage of '1010' = $|V_{ref}|(\frac{R_f}{R_3} + 0.25 \times \frac{R_f}{R_3}) = 5$

Thus, $\frac{R_f}{R_3} = 0.5$

Set-B

For the binary weighted DAC shown below, answer the following questions:



Here, $V_R = -12V$, $R_3 = R_f$ & Input bit sequence: $S_3 S_2 S_1 S_0$

(a)	CO2	Show the output voltages for all possible input cases '0XXX' in a table, where X may be 0 or 1.	[10]
(b)	CO2	What is the ratio of MSB(for input 1000) voltage & minimum non-zero output voltage?	[4]
(c)	CO2	What should be the ratio $\frac{R_3}{R_f}$, if we want the output voltage for input '1100' to be 6V?	[6]

Solution:

a)

Input	Output = $ V_R \times (B_2 \times 0.5 + B_1 \times 0.25 + B_0 \times 0.125)$
0000	0
0001	1.5
0010	3
0011	4.5
0100	6
0101	7.5
0110	9
0111	10.5

b) MSB voltage = $|V_{ref}|(1) = 12V$ Minimum non-zero voltage = $|V_{ref}|(0.125) = 1.5V$

So, ratio = 8:1

c) Output voltage of '1100' = $|V_{ref}|(\frac{R_f}{R_3} + 0.5 \times \frac{R_f}{R_3}) = 6$ Thus, $\frac{R_f}{R_3} = \frac{1}{3}$

Question 3: Schmitt Trigger [SZZ]

SET A

For the Given Schmitt Trigger Circuit, $V_H = 10V$, $V_L = -10V$, $V_{ref} = 1V$

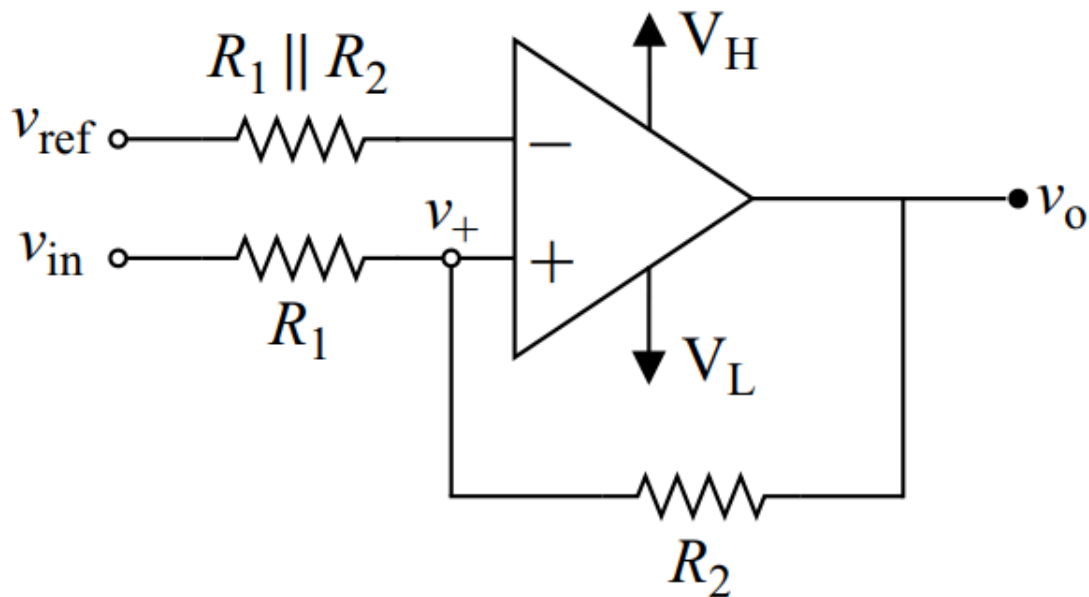
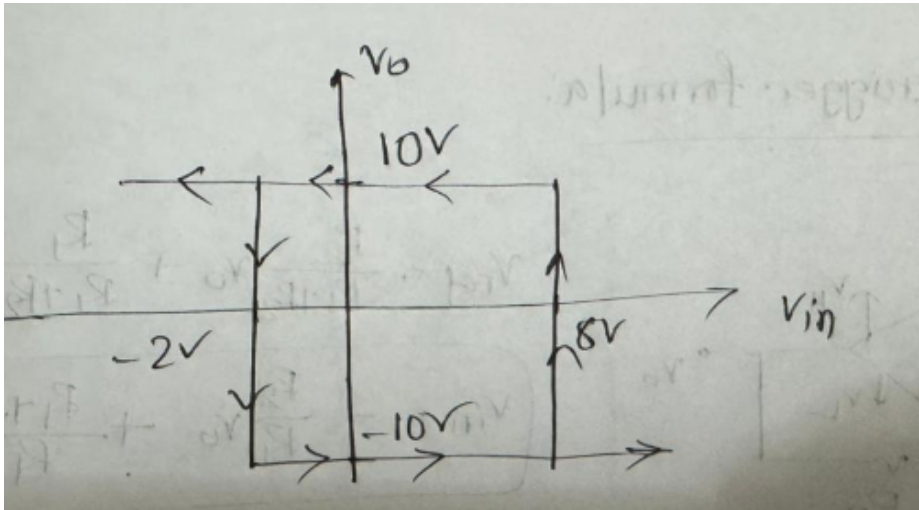


Figure: Schmitt Trigger

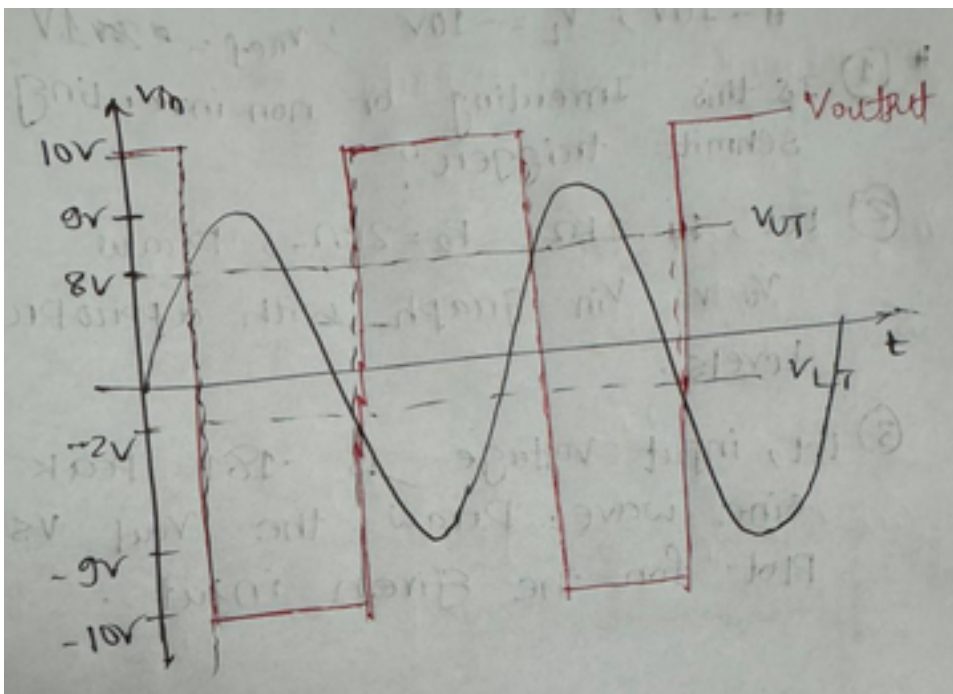
(a)	CO	Is this an Inverting or non-inverting Schmitt Trigger?	2
(b)	CO	Let $R_1 = 1 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$; Draw V_{out} vs. V_{in} graph (VTC) with appropriate labels.	10
(c)	CO	Let input voltage be 18V peak to peak sine wave. Draw V_{out} vs. time plot for the given input.	8

Solution:

a. Non-inverting.



b.



c.

SET B

For the Given Schmitt Trigger Circuit, $V_H = 10V$, $V_L = -10V$, $V_{ref} = 1V$

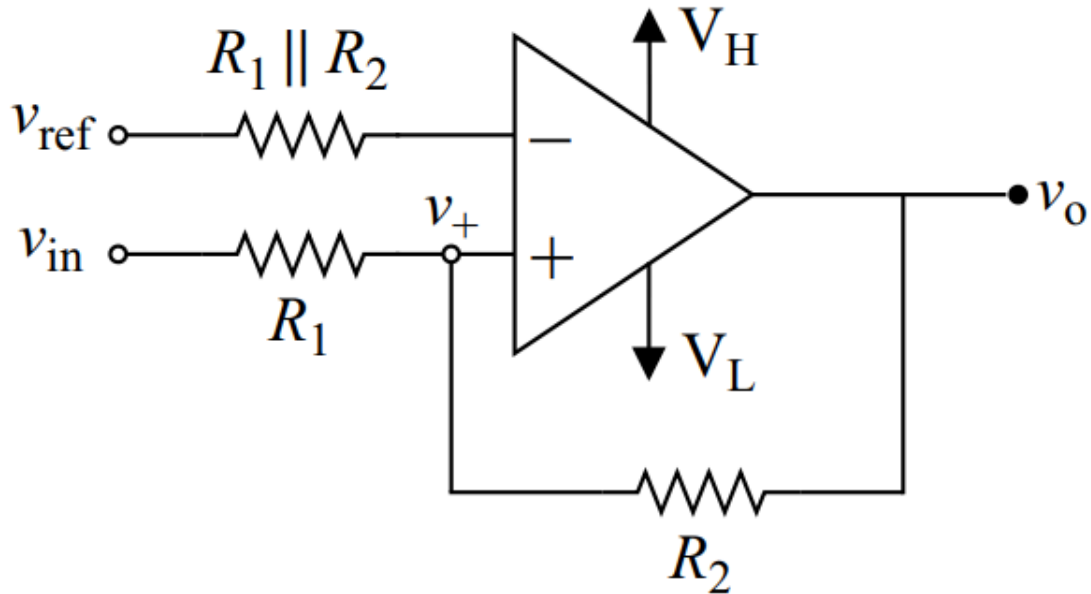


Figure: Schmitt Trigger

(a)	CO	Is this an Inverting or non-inverting Schmitt Trigger?	2
(b)	CO	Let $R_1 = 2 \text{ k}\Omega$, $R_2 = 4 \text{ k}\Omega$; Draw V_{out} vs. V_{in} graph (VTC) with appropriate labels.	10
(c)	CO	Let input voltage be 20V peak to peak sine wave. Draw V_{out} vs. time plot for the given input.	8

Question 4: Signal Generator [SDU]

(Set A)

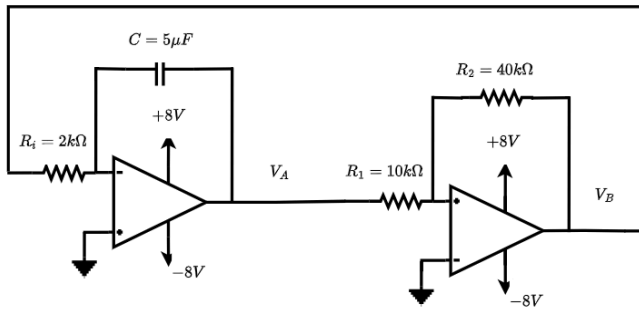


Figure 1: Triangular Wave Generator

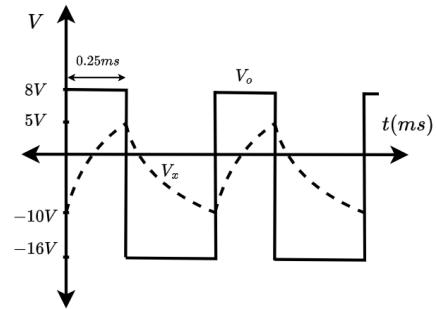


Figure 2: Square Wave Generator Output

(a)	CO3	Calculate the time period and the frequency of the triangular wave generator of Figure 1.	7
(b)	CO3	Design a square wave generator that can generate a similar output waveform given in Figure 2. Draw the designed circuit with all necessary values. (You can assume the value of the capacitor 50 nF.)	8
(c)	CO3	Find the duty cycle and frequency of the output waveform of your designed square wave generator.	5

Qstn-4 soln (Set-A)

(a) Triangular Wave

here $V_L = -V_H$, $p = \frac{40}{10} = 4$

$$\begin{cases} f = \frac{p}{4RC} = \frac{4}{4 \times 2K \times 5\mu F} = 100 \text{ Hz} \\ T = \frac{1}{f} = 10 \text{ ms} \end{cases}$$

(b) From fig-2 $\rightarrow V_H = 8V$, $V_L = -16V$, $V_{TH} = 5V$, $V_{TL} = -10V$, $T_1 = 0.25 \text{ ms}$

$$V_{TH} = \frac{R_1}{R_1 + R_2} V_H \quad \text{or,} \quad V_{TL} = \frac{R_1}{R_1 + R_2} V_L$$

$$5 = \frac{R_1}{R_1 + R_2} \times 8$$

$$\Rightarrow \frac{5}{8} = \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow \frac{5}{3} = \frac{R_1}{R_2}$$

$$\therefore R_1 = \frac{5}{3} R_2$$

$$\begin{cases} \text{let } R_2 = 30K \\ \therefore R_1 = 50K \end{cases}$$

$$-10 = \frac{R_1}{R_1 + R_2} (-16)$$

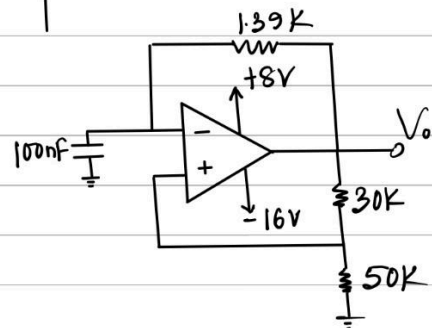
$$\frac{10}{16} = \frac{R_1}{R_1 + R_2}$$

$$R_1 = \frac{5}{3} R_2$$

$$T_1 = R_x C_x \ln \left| \frac{V_H - V_{TL}}{V_H - V_{TH}} \right|$$

$$0.25 \text{ ms} = R_x \times 100 \text{ nF} \times 1.79$$

$$\boxed{R_x = 1.39 K}$$



(c) $T_1 = 0.25 \text{ ms}$

$$T_2 = R_x C_x \ln \left| \frac{V_L - V_{TH}}{V_L - V_{TL}} \right|$$

$$= 1.396 K \times 100 \text{ nF} \times \ln \left| \frac{-16 - 5}{-16 + 10} \right|$$

$$T_2 = 0.17 \text{ ms}$$

$$\therefore \text{Period, } T = T_1 + T_2 = 0.42 \text{ ms}$$

$$\text{freq. } f = \frac{1}{T} \approx 2381 \text{ Hz}$$

$$\text{Duty cycle, } D = \frac{T_1}{T} \times 100\%$$

$$= \frac{0.25}{0.42} \times 100\%$$

$$= 59.52\%$$

(Set B)

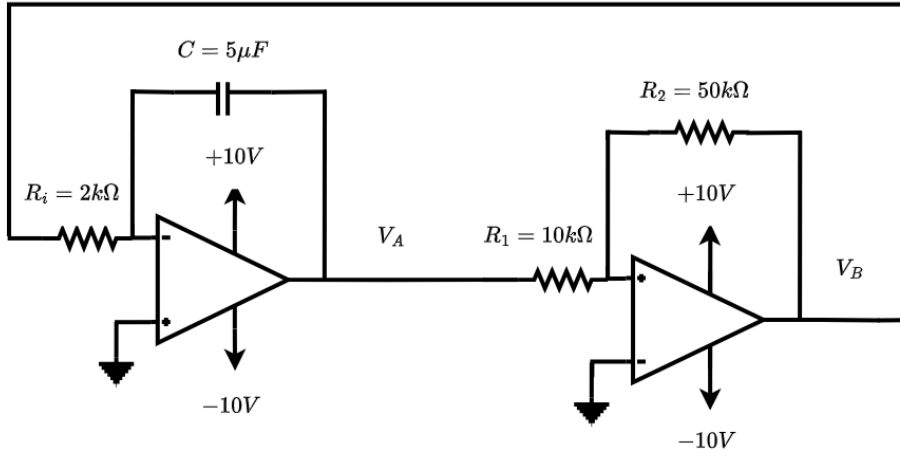
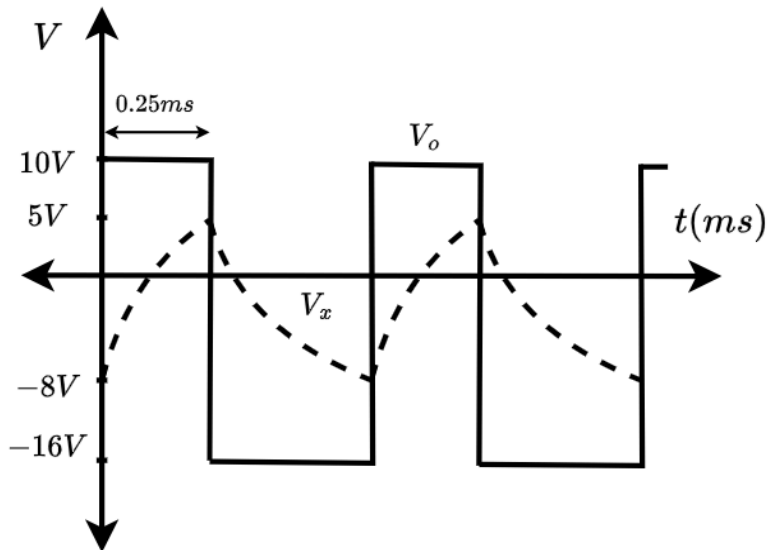


Figure 1: Triangular Wave Generator



(a)	CO3	Calculate the time period and the frequency of the triangular wave generator of Figure 1.	7
(b)	CO3	Design a square wave generator that can generate a similar output waveform given in Figure 2. Draw the designed circuit with all necessary values. (You can assume the value of the capacitor 100 nF)	8
(c)	CO3	Find the duty cycle and frequency of the output waveform of your designed square wave generator.	5

Qstn-4 soln (Set-B)

(a) Triangular Wave

here $V_L = -V_H$, $p = \frac{50}{10} = 5$

$$\begin{cases} f = \frac{p}{4R_1C} = \frac{5}{4 \times 2k \times 5\mu f} = 125 \text{ Hz} \\ T = \frac{1}{f} = 8 \text{ ms} \end{cases}$$

(b) From Fig-2 $\rightarrow V_H = 10V, V_L = -16V, V_{TH} = 5V, V_{TL} = -8V, T_1 = 0.25 \text{ ms}$

$$\begin{aligned} V_{TH} &= \frac{R_1}{R_1 + R_2} V_H \quad \text{or} \quad V_{TL} = \frac{R_1}{R_1 + R_2} V_L \\ 5 &= \frac{R_1}{R_1 + R_2} \times 10 \quad -8 = \frac{R_1}{R_1 + R_2} (-16) \\ \Rightarrow \frac{5}{10} &= \frac{R_1}{R_1 + R_2} \quad \frac{8}{16} = \frac{R_1}{R_1 + R_2} \\ \Rightarrow \frac{5}{5} &= \frac{R_1}{R_2} \quad R_1 = R_2 \end{aligned}$$

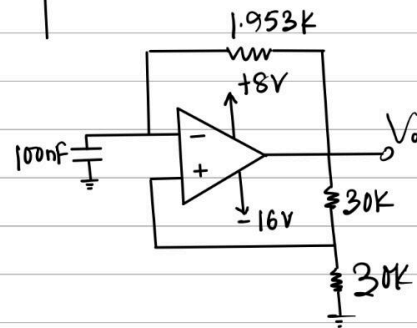
$\therefore R_1 = R_2$

Let $R_2 = 30k$
 $\therefore R_1 = 30k$

$$T_1 = R_x C_x \ln \left| \frac{V_H - V_{TL}}{V_H - V_{TH}} \right|$$

$$0.25 \text{ ms} = R_x \times 100 \text{ nF} \times 1.28$$

$$R_x = 1.953k$$



(c) $T_1 = 0.25 \text{ ms}$

$$\begin{aligned} T_2 &= R_x C_x \ln \left| \frac{V_L - V_{TH}}{V_L - V_{TL}} \right| \\ &= 1.953k \times 100 \text{ nF} \times \ln \left| \frac{-16 - 5}{-16 + 8} \right| \\ T_2 &= 0.188 \text{ ms} \end{aligned}$$

$\therefore$ Period, $T = T_1 + T_2 = 0.438 \text{ ms}$

freq, $f = \frac{1}{T} \approx 2283 \text{ Hz}$

$$\begin{aligned} \text{Duty cycle, } D &= \frac{T_1}{T} \times 100\% \\ &= \frac{0.25}{0.438} \times 100\% \\ &= 57\% \end{aligned}$$

Question 5: MOS [NUR]

(Set A)

For the NMOS used in Figure 1, Given:

$$K_n = 0.2 \text{ mA/V}^2 \quad [K_n = 1/2 \mu_n C_{ox} W/L], \quad V_{TH} = 0.6 \text{ V}$$

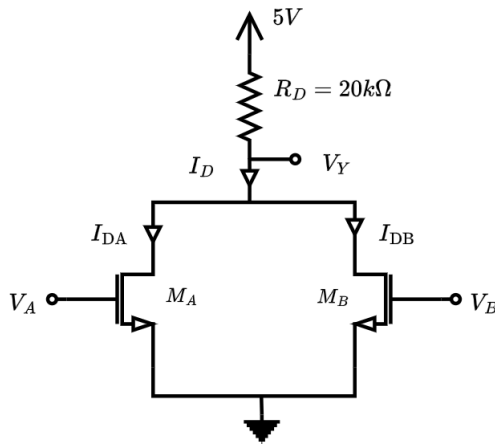


Figure 1: NMOS NOR Gate

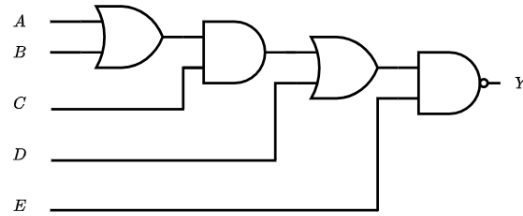


Figure 2: Logic Circuit

(a)	CO1	From the above Figure 1, determine the I_D , I_{DA} , I_{DB} and V_Y for the cases both inputs are high and both inputs are low. (Consider 5 V as logic high and 0 V as logic low)	8
(b)	CO1	Determine the power dissipations for the cases both inputs are high and both inputs are low.	4
(c)	CO1	Find the boolean expression for output Y from Figure 2 and using that expression design an equivalent CMOS logic circuit of Figure 2.	8

Qstn-5 soln (Set-A)

(a) $K_n = 0.2 \text{ mA/V}^2$, $V_{tn} = 0.6 \text{ V}$

$V_A = V_B = V_{G_S}$

$V_{D_S} = V_Y$

* Case-I $\rightarrow V_A = 5 \text{ V}$, $V_B = 5 \text{ V}$

$M_A, M_B \rightarrow$ triode region

$I_D = I_{D_A} + I_{D_B}$

$\Rightarrow \frac{5 - V_Y}{20K} = K_n \{ 2(V_{G_S} - V_{tn})V_{D_S} - V_{D_S}^2 \} \times 2$

$\Rightarrow (5 - V_Y) = 20 \times 0.2 \times \{ 2(5 - 0.6)V_Y - V_Y^2 \} \times 2$

$\Rightarrow 5 - V_Y = 8 (8.8V_Y - V_Y^2)$

$\Rightarrow 5 - V_Y = 70.4V_Y - 8V_Y^2$

$\Rightarrow 8V_Y^2 - 71.4V_Y + 5 = 0$

Solving $\rightarrow V_Y = 8.854, 0.07 \text{ V}$

$\therefore V_Y = 0.07 \text{ V}$

$I_D = \frac{5 - 0.07}{20K} = 0.2465 \text{ mA}$

$I_{D_A} = I_{D_B} = \frac{I_D}{2} = 0.123 \text{ mA}$

* Case-II : $V_A = 0 \text{ V}$, $V_B = 0 \text{ V}$
 $M_A, M_B \rightarrow$ Cutoff

$I_D = I_{D_A} = I_{D_B} = 0 \text{ mA}$

$I_D = 0$

$\frac{V_{DD} - V_Y}{20} = 0 \therefore V_Y = V_{DD}$

$V_Y = 5 \text{ V}$

(b) Power Consumption

* Case-I

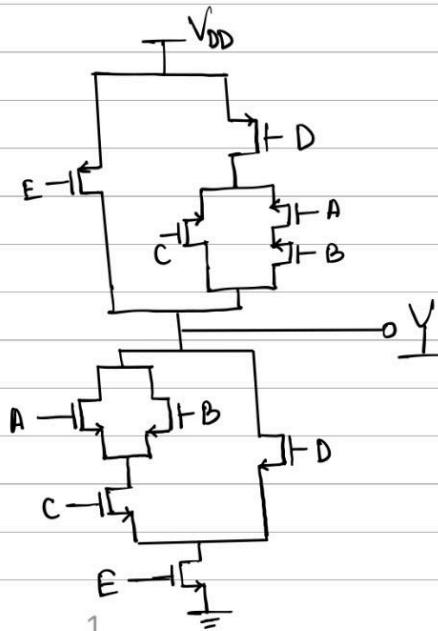
$P_1 = (V_{DD} - 0) I_D$
 $= (5 - 0) \times 0.2465$
 $= 1.23 \text{ mW}$

* Case-II

$P_2 = (V_{DD} - 0) I_D$
 $= 0 \text{ mW}$

(c)

$Y = ((A+B)C+D)E$



(Set B)

For the NMOS used in Figure 1, Given:

$$K_n = 0.2 \text{ mA/V}^2 \quad [K_n = 1/2 \mu_n C_{ox} W/L], \quad V_{TH} = 0.6 \text{ V}$$

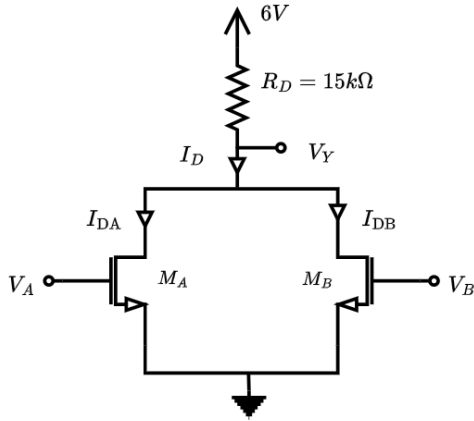


Figure 1: NMOS NOR Gate

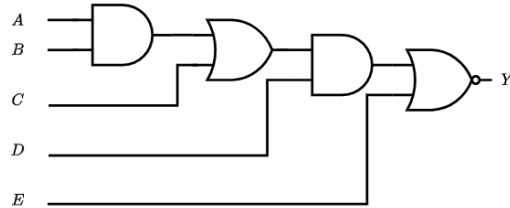


Figure 2: Logic Circuit

(a)	CO1	From the above Figure 1, determine the I_D , I_{DA} , I_{DB} and V_Y for the cases both inputs are high and both inputs are low. (Consider 6 V as logic high and 0 V as logic low)	8
(b)	CO1	Determine the power dissipations for the cases both inputs are high and both inputs are low.	4
(c)	CO1	Find the boolean expression for output Y from Figure 2 and using that expression design an equivalent CMOS logic circuit of Figure 2.	8

Qstn-5 soln (Set-B)

(a) $K_n = 0.2 \text{ mA/V}^2$, $V_{tn} = 0.6 \text{ V}$

$V_A = V_B = V_{GS}$

$V_{DS} = V_Y$

* Case-I $\rightarrow V_A = 6 \text{ V}$, $V_B = 6 \text{ V}$

$M_A, M_B \rightarrow$ triode region

$I_D = I_{DA} + I_{DB}$

$\Rightarrow \frac{6 - V_Y}{15 \text{ k}} = K_n \{ 2(V_{GS} - V_{tn})V_{DS} - V_{DS}^2 \} \times 2$

$\Rightarrow (6 - V_Y) = 15 \times 0.2 \times \{ 2(6 - 0.6)V_Y - V_Y^2 \} \times 2$

$\Rightarrow 6 - V_Y = 6 (10.8 V_Y - V_Y^2)$

$\Rightarrow 6 - V_Y = 64.8 V_Y - 6 V_Y^2$

$\Rightarrow 6 V_Y^2 - 65.8 V_Y + 6 = 0$

Solving $\rightarrow V_Y = 10.87, 0.091 \text{ V}$

$\therefore V_Y = 0.091 \text{ V}$

$I_D = \frac{6 - 0.091}{15} = 0.393 \text{ mA}$

$I_{DA} = I_{DB} = \frac{I_D}{2} = 0.196 \text{ mA}$

* Case-II: $V_A = 0 \text{ V}$, $V_B = 0 \text{ V}$
 $M_A, M_B \rightarrow$ Cutoff

$I_D = I_{DA} = I_{DB} = 0 \text{ mA}$

$I_D = 0$

$\frac{V_{DD} - V_Y}{15} = 0 \therefore V_Y = V_{DD}$
 $V_Y = 6 \text{ V}$

(b) Power Consumption

* Case-I

$P_1 = (V_{DD} - 0) I_D$
 $= (6 - 0) \times 0.393$
 $= 2.358 \text{ mW}$

* Case-II

$P_2 = (V_{DD} - 0) I_D$
 $= 0 \text{ mW}$

(c) $Y = (AB + C)D + E$

